

Optimization engine technical report

TR · MILP · Customer-scale CVRPTW · Public Solomon benchmark

Measured results at a glance

EXPERIMENT	MEASURED RESULT	TECHNICAL SIGNIFICANCE
Warehouse MILP · 316,814 variables	cuOpt 15.4 s total; CPU comparators 21.4–44.1 s	1.39–2.86× faster at the tested scale; reported same optimum and zero violations.
GS Retail CVRPTW · 537–586 orders	cuOpt 2,610 km at 60 s; OR-Tools 2,815 km	7.28% lower aggregate distance versus OR-Tools on three instances; +0.83% reported gap.
Solomon · 56 × 100–customer instances	RouteOpt +0.08% gap, 38/56 wins; cuOpt +1.76%, 13/56	Complementary engines: solver selection depends on problem class, scale and budget.

Report basis

Issuer: Omelet. Prepared 17 September 2026 from the final-pitch experimental results. Each study states the metric, comparison, test setup, results and the date/owner information disclosed in the public appendix. This is a company technical report, not an independent rerun or a claim of completed production integration. Customer-level inputs remain confidential.

Product implementation

[Watch the product demonstrations](#) to see OI Factory, Studio, Foundry and industrial application interfaces. [Integration status](#) distinguishes current software demonstrations, solver experiments and the NVIDIA roadmap.

cuOpt benchmark: warehouse MILP

Result: cuOpt reaches the reported optimum in **15.4 seconds total** at 316,814 variables: **1.39–2.86× faster** than the compared CPU solvers on this large instance.

Metric	Wall-clock time to optimality; solve time and total time reported separately.
Problem	316,814 variables and 240,120 constraints; warehouse planning/slotting.
Comparison	cuOpt 26.08 on one RTX 4090 versus SCIP 9.0, CP-SAT 9.11 and CBC 2.10 on CPU, 64 threads.
Setup	Xeon Gold 6326 x2; RTX 4090, 24 GiB (2.0 GiB used as reported); CUDA 13.0; Ubuntu 24.04. Three-run median; solutions independently re-simulated as reported.
Date / reporting entity	Report: Omelet, 17 September 2026. The public appendix does not specify the run date or individual experiment owner.

SOLVER	HARDWARE	SOLVE TIME	TOTAL TIME	CPU TOTAL / CUOPT TOTAL
NVIDIA cuOpt 26.08	1 GPU	13.6 s	15.4 s	1.00×
SCIP 9.0	CPU, 64T	18.1 s	21.4 s	1.39×
CP-SAT 9.11	CPU, 64T	15.0 s	22.6 s	1.47×
CBC 2.10	CPU, 64T	40.8 s	44.1 s	2.86×

Feasibility and scaling

The appendix reports all four solvers reaching the same optimum: 1,139 pallet positions, with 0 constraint violations on re-simulation and a reported 24.3% reduction in peak space. The operational baseline behind the peak-space reduction is not specified.

For an increase from 62K to 317K variables (5.1×), reported time growth is 1.7× for cuOpt, 6.3× for CP-SAT, 7.9× for SCIP and 9.4× for CBC. At smaller, single-warehouse scale, SCIP is reported at 0.2–2.7 seconds. GPU benefit is scale-dependent.

Technical interpretation

The result demonstrates a GPU advantage at the tested problem scale while preserving the reported optimum and feasibility. Reproduction requires the same model instance, solver settings and tolerances; this public report provides aggregate results rather than a downloadable run package.

[Open appendix A2: full results](#) · [Read the consolidated technical report](#)

cuOpt benchmark: GS Retail routing

Result: cuOpt returns **2,610 km at 60 seconds**, compared with 2,815 km for OR-Tools: approximately **7.3% shorter aggregate distance** on the reported three-instance comparison.

Metric	Route distance (km) and per-instance gap to the best result among compared solvers; 10- and 60-second budgets.
Problem	Three GS Retail instances; A3 replenishment scenario; 537–586 orders.
Setup	Full-results appendix: one RTX A6000, August 2026, drive-distance objective and 55 km/h. Summary appendix also mentions RTX 4090 / A6000. Repeated runs and re-validation are reported; per-run device mapping/repetition count are not itemized in the public appendix.
Date / reporting entity	Experiment: August 2026. Report issuer: Omelet. The public appendix does not name the individual experiment owner.

METHOD	10 S: KM	10 S: GAP	60 S: KM	60 S: GAP
NVIDIA cuOpt	2,684	+1.09%	2,610	+0.83%
Omelet RouteOpt	2,676	+1.37%	2,622	+1.70%
RouteOpt, ALNS configuration	2,984	+12.22%	2,766	+7.16%
OR-Tools	3,000	+12.79%	2,815	+9.50%
LKH-3	No feasible solution	at 1,830 s	≈30× the 60 s budget	—

How to interpret the results

Gap uses a moving, per-instance best-across-solvers reference. Compare distance across time budgets, not the gap percentages as if they shared a fixed global optimum. Aggregated km and mean per-instance gap may rank differently; raw per-instance records are needed to reconstruct the averages.

The appendix reports 2/3 cuOpt wins at both budgets and run-to-run variation: cuOpt ±1.24 / ±0.50 percentage points and RouteOpt-ALNS ±0.87 / ±1.27 percentage points (10 / 60 seconds). The deck describes the frontier difference as within noise. A formal significance/equivalence test and comparable variation for primary RouteOpt are not supplied; no statistical-tie conclusion is independently verified here.

Technical interpretation

Distance reduction versus OR-Tools is calculated as $(2,815 - 2,610) / 2,815 = 7.28\%$. This is a solver comparison on the tested instances, not a customer cost-savings measurement. Customer orders and routes remain confidential.

[Open appendix A2: full results](#) · [Read the consolidated technical report](#)

Routing benchmark: Solomon instances

Result: Omelet RouteOpt records a **+0.08% reported gap and 38/56 wins**. cuOpt records +1.76% and 13/56 wins, illustrating why the platform retains complementary solver backends.

Metric	Reported gap, instance win count and runtime.
Baseline / comparison	Omelet RouteOpt, cuOpt, RouteOpt-ALNS, LKH-3 and OR-Tools.
Setup	Public Solomon CVRPTW set; 56 × 100-customer instances; nominal 10-second budget. The appendix describes same-instance, same-machine comparisons; a separate Solomon machine/run configuration is not listed.
Gap reference	Best compared solver per instance, as defined in the full-results appendix; not an external best-known-solution database.
Date / reporting entity	Report: Omelet, 17 September 2026. The public appendix does not separately state the Solomon run date or individual experiment owner.

METHOD	HARDWARE	REPORTED GAP	WINS / 56	REPORTED TIME
Omelet RouteOpt	CPU	+0.08%	38	10.0 s
NVIDIA cuOpt	CPU + GPU	+1.76%	13	9.7 s
RouteOpt, ALNS configuration	CPU	+2.53%	3	8.1 s
LKH-3	CPU	+3.41%	2	47.6 s
OR-Tools	CPU	+4.56%	0	10.0 s

Time/feasibility exception: LKH-3 is reported feasible on 52/56 instances and at 47.6 seconds. This row is not an equal 10-second-budget result. Handling of infeasible cases in the aggregate gap is not specified.

Technical interpretation

Solver choice depends on the problem class and time budget: learned/hybrid RouteOpt leads this small-instance set, while the separate large-scale routing and MILP experiments show cuOpt strengths. Aggregate tables do not replace per-instance run records for reproduction.

[Open appendix A2: full results](#) · [Read the consolidated technical report](#)